

Technical Document #96: Deep Learning - Comprehensive Overview

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Generative adversarial networks (GANs) consist of two neural networks competing against each other. The generator creates synthetic data while the discriminator evaluates authenticity.

Transfer learning is a technique where a model trained on one task is re-purposed on a second related task. It has revolutionized computer vision and NLP.

Attention mechanisms allow models to focus on different parts of the input when producing output. They have become essential in sequence-to-sequence models.

Batch normalization normalizes the inputs of each layer, reducing internal covariate shift. It allows higher learning rates and acts as a regularizer.

Recurrent neural networks (RNNs) are designed to process sequential data. They maintain hidden states that capture information about previous elements in the sequence.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies.
- Residual connections enable training of very deep neural networks by providing shortcuts for gradient flow.
- Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions.